# From Mechanistic Interpretability to Mechanistic Biology: Training, Evaluating, and Interpreting Sparse Autoencoders on Protein Language Models

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## Abstract

Protein language models (pLMs) are powerful predictors of protein structure and function, learning through unsupervised training on millions of protein sequences. pLMs are thought to capture common motifs in protein sequences, but the specifics of pLM features are not well understood. Identifying these features would not only shed light on how pLMs work, but potentially uncover novel protein biology—studying the model to study the biology. Motivated by this, we train sparse autoencoders (SAEs) on the residual stream of a pLM, ESM-2. By characterizing SAE features, we determine that pLMs use a combination of generic features and family-specific features to represent a protein. In addition, we demonstrate how known sequence determinants of properties such as thermostability and subcellular localization can be identified by linear probing of SAE features. For predictive features without known functional associations, we hypothesize their role in unknown mechanisms and provide visualization tools to aid their interpretation. Our study gives a better understanding of the limitations of pLMs, and demonstrates how SAE features can be used to help generate hypotheses for biological mechanisms. We release our code, model weights and feature visualizer.

## 1. Introduction

Protein language models (pLMs) are language models trained on large datasets of protein sequences. In the process of minimizing loss on their pre-training task–usually masked token prediction–they learn representations of proteins useful for downstream tasks including protein structure (Lin et al., 2023) and function (Rao et al., 2019; Notin et al., 2022) prediction. For this reason, pLMs have become a growing component of the protein biologist's toolkit.

As these models are widely adopted, a number of studies have investigated their inner workings to understand their limitations. Previous work suggests pLMs do not learn the biophysics of proteins, but rather store common sequence motifs and contacts (Zhang et al., 2024). For many downstream tasks, pLM performance is driven by features learned in early layers and does not scale well with pre-training (Li et al., 2024). Yet exactly what pLMs have learned about proteins remains unknown.

Recent work in mechanistic interpretability points to a path forward. Sparse autoencoders (SAEs) have successfully extracted interpretable features from large language models (LLMs) like GPT-4 and Claude (Gao et al., 2024; Templeton et al., 2024). SAEs decompose model activations into a sparse, high-dimensional representation where individual latent dimensions often have interpretable activation patterns. These features provide a window into the inner workings of these complex models. Training SAEs on pLMs can serve an additional purpose. Unlike natural languages, which we intuitively understand, the language of proteins is more cryptic. SAE features from pLMs could potentially unveil patterns in protein sequences that we have not yet discovered.

In this work, we train SAEs on the residual stream of ESM-2, a widely used pLM (Lin et al., 2022) (Figure 1a). To interpret the SAE latents, we develop InterProt, a tool that visualizes latent activations on protein sequences and structures. We find that some features are highly interpretable, corresponding to a range of concepts from secondary structure elements to entire domains, as reported contemporaneously by Simon & Zou (2024). We conduct a human evaluation study and find that SAE latents are consistently rated as interpretable, in contrast to the ESM baseline. Furthermore, we find that many features activate highly only on specific protein families, suggesting that pLMs rely on internal representations of protein families to perform their pre-training task. We introduce a method to categorize SAE latents based on their family specificity and activation pat-

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*Proceedings of the $42^{nd}$ International Conference on Machine Learning*, Vancouver, Canada, PMLR 267, 2025. 

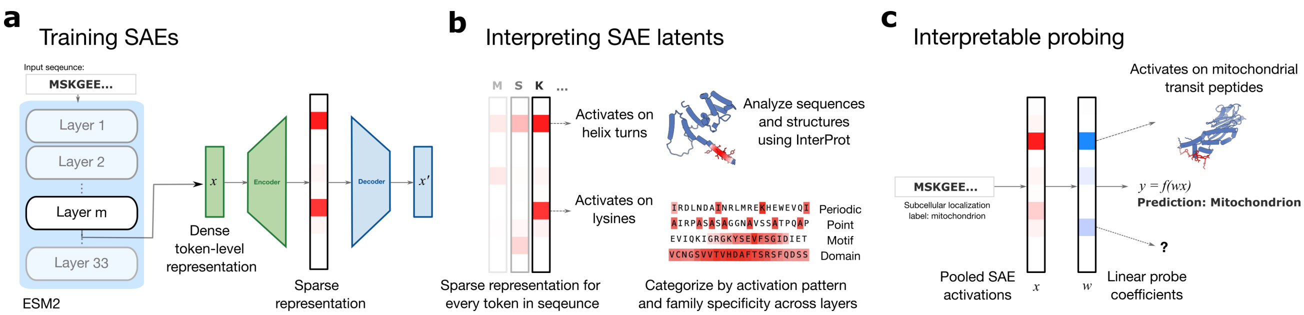


*Figure 1.* Overview of our paper. **a**. We train SAEs on the output of ESM intermediate layers. **b**. We interpret SAE latents using our latent visualizer InterProt and categorize features based on their activation pattern and family specificity. **c**. By interpreting the weights of linear models trained on SAE latents, we show how features which correspond to known sequence determinants can be identified.

tern (Figure 1b).

Having developed a framework for understanding features, we explore the impact of SAE hyperparameters and ESM layers. We find that increasing the sparsity imposed on SAE latents increases the number of features that are family-specific as does increasing the expansion factor. The number of family-specific features also varies substantially by ESM layer, with middle layers containing the most family-specific features.

With the goal of mapping SAE features to important protein properties, we train linear probes on SAE features and analyze the most predictive features. Our method identifies features corresponding to known sequence determinants for properties such as thermostability and subcellular localization. These results demonstrate that SAEs can help us not only understand pLMs, but also the data on which they are trained. By making pLM representations interpretable, we envision that SAE features may help generate hypotheses for unknown biological mechanisms (Figure 1c).

## 2. Related Works

### 2.1. Interpreting protein language models

Previous evaluations of pLM representations on downstream tasks provide evidence of features that pLMs may be learning (Rao et al., 2019; Dallago et al., 2021; Detlefsen et al., 2022; Li et al., 2024). In particular, Li et al. 2024 performs a comprehensive analysis of ESM performance on a range of downstream tasks across different model sizes and layers. Of the tasks they tested, they find that only structure-related prediction tasks scales with model size, and that most tasks use "low-level features" learned early in pre-training. A key limitation of using downstream tasks to understand the features learned by pLMs is that features cannot be discovered without supervision. In contrast, SAEs can uncover features in a more unsupervised manner.

pLMs seem to use a notion of sequence homology and recurring protein motifs. Using influence functions, Gordon et al. (2024) finds that pLM-derived sequence likelihoods are driven by homologous protein training data. Zhang et al. (2024) propose that pLMs store common motifs in proteins and the pairwise contacts between them. They arrive at this proposition by analyzing what residue segments must be unmasked to recover a sequence contact (via the "categorical Jacobian").

Many studies have explored how pLMs work mechanistically as well. Analyses of attention matrices in pLMs have been shown to resemble pairwise contact maps (Vig et al., 2021; Rao et al., 2021; Lin et al., 2023), and contain information such as binding sites (Vig et al., 2021) and allosteric sites (Kannan et al., 2024; Dong et al., 2024). They have been comprehensively mapped to Gene Ontology terms (Chen et al., 2025).

Concurrent to our work, Simon & Zou (2024) trained SAEs on ESM-2 and presented methods to analyze their latents: visualizing them, evaluating them against Swiss-Prot annotations, and interpreting them with a language model. Our work builds on this foundation with a different SAE architecture and evaluation methods that focus on coevolution, human preference, and downstream tasks (subsection A.3).

### 2.2. Dictionary learning for feature extraction from models trained on biological data

Following the success of extracting interpretable features from LLMs, a number of works have applied sparse dictionary learning techniques to models trained on scientific data. Donhauser et al. (2024) evaluates the possibility of extracting features from models trained on microscopy data. Interpretable features have also been uncovered from single cell foundation models (Schuster, 2024; Adam Green, 2024). Associating the extracted features with functional labels via probing, however, has not yet been explored in

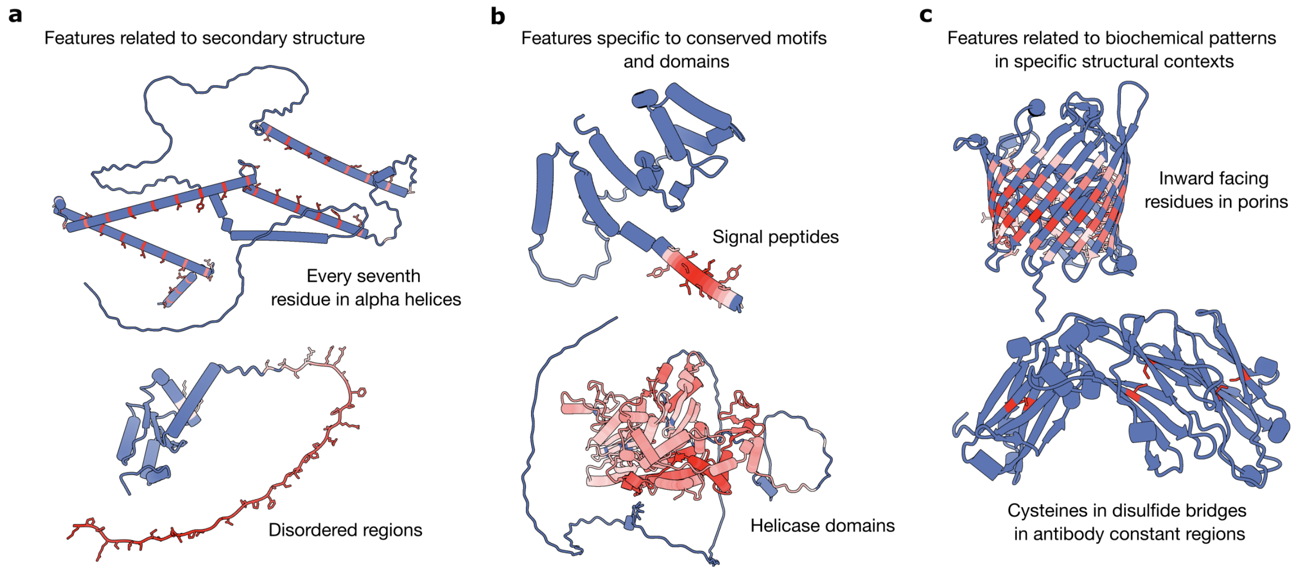


*Figure 2.* Examples of SAE features. We find features related to secondary structure (a), conserved motifs and domains (b), and biochemical patterns in specific structural contexts (c). The structures are colored according to activation (red: activation, blue: no activation).

this context.

# 3. Background and Methods

## 3.1. Sparse autoencoders (SAEs)

An SAE is an autoencoder designed to learn efficient and meaningful representations by enforcing sparsity constraints on encoder output. We use SAEs with TopK activation to enforce sparsity, as introduced in Gao et al. (2024). This activation function only allows the $k$ largest latents to be non-zero. We choose TopK SAEs due to their improved reconstruction at a given level of sparsity compared to other techniques, and because it allows us to directly set the L0-norm of the encoding using $k$. We note that improved reconstruction may come at the cost of increased feature absorption (Karvonen et al., 2024).

The encoder and decoder are defined as:

$$z = \mathrm{TopK}\left(W_{\mathrm{enc}}\left(x - b_{\mathrm{pre}}\right)\right)$$
$$\hat{x} = W_{\mathrm{dec}} z + b_{\mathrm{pre}}$$

where $W_{\mathrm{enc}}$ projects the residual stream into the SAE latent space and $W_{\mathrm{dec}}$ the reverse. TopK zeros all latents that are not in the top $k$. The loss is simply the reconstruction mean squared error $\mathcal{L} = \|x - \hat{x}\|_2^2$. We train our TopK SAEs on 1 million random sequences under 1022 residues from UniRef50 (Suzek et al., 2007).

Following Lieberum et al. (2024), we refer to the hidden dimensions of the SAE as "latents" to clearly distinguish them from the underlying conceptual features of the model. This is in contrast to previous works which use the word feature to refer to both (Bricken et al., 2024).

## 3.2. Downstream tasks

**Secondary structure.** The secondary structure dataset from TAPE (Rao et al., 2019) contains residue-level labels in 3 classes: alpha helix, beta strand, or other. To prevent data leakage, sequences are divided into training and test sets based on a threshold of 25% sequence identity.

**Subcellular localization.** Subcellular localization is a protein-level label for where a protein is localized within a cell. The dataset from Almagro Armenteros et al. (2017) consists of UniProt-sourced eukaryotic proteins, with stringent homology-based train-test splits.

**Thermostability.** The thermostability dataset from the Melteome Atlas (Jarzab et al., 2020) measures the melting temperature of 48,000 proteins across 13 species. We use the train-test split from FLIP (Dallago et al., 2021) containing all sequences clustered at 25% sequence identity.

**Mammalian cell expression.** The dataset curated by Masson et al. (2022) contains expression data of 2,165 proteins from the human secretome in Chinese hamster ovary (CHO) cells, commonly used mammalian hosts for therapeutic protein production. We use this dataset as a binary classification task to determine whether a protein can be successfully expressed in CHO cells. We split proteins into training and